Σ^0 Production in the NuMI Beam at MicroBooNE: Analysis Note

J. Nowak, N. Patel

November 1, 2026

Abstract

Working note for the MicroBooNE search for Σ^0 production in $\bar{\nu}_\mu$ charged-current interactions in the NuMI beam, $\Sigma^0 \to \Lambda\gamma$, $\Lambda \to p\pi^-$. The results are for the reverse-horn-current (RHC) exposure of 1.15×10^{21} POT; the forward-horn-current (FHC) exposure, whose wrong-sign $\bar{\nu}_\mu$ adds almost as much signal again, is to be included once its signal samples are complete. Two selections are maintained. The main one runs on standard PeLEE ntuples and ends in a cross-validated gradient BDT: $S = 2.12$ NuWro or 8.86 GENIE signal events on $B = 345$. The second follows the MicroBooNE Λ CCQE analysis on HyperonProduction ntuples, fitting the Λ decay vertex, testing the connectedness of the raw wire activity and requiring a proton-like proton candidate: $S = 0.126$ NuWro or 0.516 GENIE on $B = 2.5$. Neither can observe Σ^0 at the predicted rate; the deliverable is an upper limit. The best median expected 90% CL limit is about 4.1× the GENIE rate and 17× the NuWro rate, about 210 in-FV Σ^0 events. The two generators differ by a factor of four in the predicted rate, almost entirely because NuWro includes hyperon final-state interactions. GENIE's $\Delta S = 1$ quasi-elastic cross section was found to be wrong in its hyperon-mass terms; every GENIE sample is corrected by event weights, and the GENIE signal samples are being regenerated with a fixed GENIE. The PeLEE chain gives the better limit up to a background systematic of about 20% and the Λ-style chain above it, so the PeLEE background systematic decides which chain leads and is the next piece of work.

Contents

1 Status **2**

2 Signal, samples and signal models **3**
2.1 Signal definition and fiducial volume . . . 3
2.2 Samples . . . 3
2.3 The two signal models . . . 3
2.4 GENIE's hyperon cross section and the correcting weights . . . 4
2.5 FHC . . . 5

3 The PeLEE selection **6**
3.1 Candidates and cut chain . . . 6
3.2 BDT . . . 6
3.3 Results . . . 6
3.4 GENIE signal . . . 6
3.5 Detector variations . . . 7

4 The Λ-style selection **8**
4.1 Operating point and results . . . 8
4.2 Beam-off background . . . 8
4.3 What the remaining background is . . . 9
4.4 Selection optimisation . . . 10

5 Expected sensitivity **11**
5.1 Method . . . 11
5.2 Results . . . 11

6 Open items and plan **12**

A GENIE's hyperon cross section in detail **13**
A.1 The fixed GENIE . . . 14

B Samples: history and validation **14**

C PeLEE BDT development **15**

D What limits the PeLEE purity **16**

E Reconstruction study on HyperonProduction ntuples **16**

F Λ-style chain: development and checks **17**

G The hypana BDT chain **18**

H Methods kept for later **18**

I Change log **19**

1 Status

The analysis sets a limit on Σ^0 production in $\bar{\nu}_\mu$ charged-current interactions,

$$\bar{\nu}_\mu + \mathrm{Ar} \to \mu^+ + \Sigma^0 + X, \qquad \Sigma^0 \to \Lambda\gamma, \qquad \Lambda \to p\,\pi^-, \tag{1}$$

so the reconstructed final state is a muon, a proton, a pion and a photon of about 75 MeV. All yields are scaled to the RHC data exposure, 1.1526×10^{21} POT.

Two selections exist, on two ntuple formats, and they are fully independent (Table 1). The PeLEE chain (`Sigma/pelee/SigmaSelection.C`, Sec. 3) is the analysis: it is the only one that can see data, EXT and dirt. The Λ-style chain (`SigmaLambdaSel.C`, `IslandProcess.C`, Sec. 4) needs the raw wire activity, which only the HyperonProduction ("hypana") format stores; that format exists for MC and EXT but not for data or dirt.

Table 1: The two selections at their operating points, RHC. Signal yields carry every weight: the NuWro flux weight, PPFX on GENIE, the GENIE hyperon cross-section correction and the Σ^0 polarisation (Sec. 2). μ_{UL} is the median expected 90% CL limit on the rate in units of the model, at a 15% background systematic (Sec. 5).

	PeLEE BDT	Λ-style
ntuple format	PeLEE	HyperonProduction
data, EXT, dirt	all three	EXT only (factorised)
final discriminant	BDTG > 0.779	island test on ≥ 2 planes, proton PID
in-FV signal, NuWro / GENIE	12.55 / 49.27	
S, NuWro (with hyperon FSI)	2.117	0.126
S, GENIE (no hyperon FSI)	8.86	0.516
B	344.9	2.48
purity, NuWro / GENIE	0.61% / 2.5%	4.8% / 17%
μ_{UL} ($f = 15\%$), NuWro / GENIE	44 / 10.5 (cut re-optimised: 30 / 7.0)	33 / 8.1

The main facts.

- **The result will be a limit.** The best median discovery significance at the predicted rate is 0.45σ (GENIE) and 0.1σ (NuWro). The PeLEE ntuples carry the multisim weights the background covariance needs (`weightsGenie`, `weightsPPFX`, `weightsReint`).
- **The two signal models differ by a factor of four in rate** (49.3 against 12.6 in-FV Σ^0), almost all of it hyperon final-state interactions, which NuWro models and GENIE does not (Sec. 2.3). Their production rates agree to about 10% once GENIE's cross section is corrected. Every result is quoted under both.
- **GENIE's Σ^0 cross section was 1.85× too high** and its Λ 0.8× too low (Sec. 2.4, App. A). All GENIE samples carry a correcting weight; the GENIE signal samples are being regenerated with a fixed GENIE (Sec. 6).
- **Which chain gives the better limit depends on the PeLEE background systematic**: the PeLEE chain below $f \approx 20\%$, the Λ-style chain above. That systematic drives the PeLEE result (its limit grows 1.7× from $f = 0$ to 15% with the cut re-optimised), while the Λ-style limit is Poisson-limited and almost flat in f.
- **FHC adds almost as much signal as RHC** (0.90× per POT, 47 expected GENIE Σ^0 in 1.10×10^{21} FHC POT against 54 in RHC); its background per POT will be higher (Sec. 2.5).

2 Signal, samples and signal models

2.1 Signal definition and fiducial volume

$\bar{\nu}_\mu$ charged current with a Σ^0 in the primary final state, a true muon momentum above 0.15 GeV/c (the CC1μ1π threshold; it removes 4.2% of signal muons), no other meson at the primary vertex, and the true vertex inside the fiducial volume: $3.00 < x < 253.35$, $-112.53 < y < 114.47$, $3.1 < z < 1033.0$ cm, excluding $675.1 < z < 775.1$ cm. This is looser than the CC1μ1π box; the signal is rare enough that acceptance is worth more than vertex quality. The PeLEE truth record stops at the primary vertex, so no proton, pion or photon threshold can be imposed; the meson veto removes nothing from the NuWro sample and makes the GENIE QE sample complete (App. B).

2.2 Samples

Per-event weights:

- **NuMI flux.** NuWro's flux files came from the old NuMI flux (g4numi v6, bugged beamline geometry), while the overlay and data are on the current `g4_10_4` flux. Every NuWro event carries the official New/Old ratio weight in $(E_\nu, \theta_{\text{beam}})$, mean 0.677 and flat to 0.1% for the signal subset; 49.9k effective events of 58.5k. The RHC dirt is on the old flux too and takes the same map.
- **PPFX.** The overlay carries `ppfx_cv`; the GENIE QE sample, made from generator files, takes the overlay's $\langle$`ppfx_cv`$\rangle$ in true E_ν (`PpfxMap.C`), 0.934 on the signal. Hypana MC carries no PPFX weight; the Λ-style GENIE yield is multiplied by 0.934.
- **GENIE hyperon cross section and Σ^0 polarisation** (Sec. 2.4).

2.3 The two signal models

NuWro and GENIE predict very different Σ^0 yields (Table 3). The difference has two parts: the production cross section, where GENIE was wrong and is corrected, and hyperon final-state interactions, a genuine model difference.

Table 2: Samples, scaled to 1.1526×10^{21} POT. History and validation of each sample are in App. B.

format	sample	content	scale
PeLEE	Σ^0 signal, other hyperons	NuWro run3b overlay, 249 832 events, 1.3394×10^{24} POT	8.61×10^{-4}
	GENIE Σ^0, fake data	GENIE Σ^0 QE overlay, 250 555 events, 1.2300×10^{24} POT	9.37×10^{-4}
	ν background	RHC overlay Runs 1, 2, 3, 4a, 4b, 4c, 1.5463×10^{22} POT	0.0745
	dirt	RHC Runs 1, 2b, 3b, 4a, 4b, 1.7409×10^{21} POT	0.662
	beam-off (EXT)	run3b, 3.265×10^7 gates	0.750
hypana	Σ^0 signal (GENIE)	`sig_genie_qe.list`, the QE sample above, 2506 files	9.37×10^{-4}
	Σ^0 signal (NuWro)	the run3b production above	8.61×10^{-4}
	ν background	`bkg_overlay.list`, 14 164 files, 7.0488×10^{21} POT	0.1635
	beam-off (EXT)	16 570 events from 3 349 run3b files, 162 912 gates	150.25
	dirt, data	none	

Table 3: In-FV $\bar{\nu}_\mu$ CC Σ^0 per 1.1526×10^{21} POT. "Produced" counts the Σ^0 made at the vertex, "surviving" those that leave the nucleus as Σ^0.

	produced	surviving
GENIE, Pais cross section (as generated)	83.3	83.3
GENIE, cross section corrected	55.8	55.8
NuWro (flux-weighted)	53.6	12.8

Hyperon final-state interactions. NuWro transports the hyperon through the nucleus; GENIE's INTRANUKE does not re-interact hyperons, so for GENIE produced equals surviving. In the NuWro sample only 25.3% of $\bar{\nu}_\mu$ CC events still carry a Σ^0, 27.9% carry a Λ ($\Sigma^0 N \to \Lambda N$), 10% a $\Sigma^\pm$, and 37.3% no strange particle at all (hyperons below the binding threshold are dropped; the build used, `nuwro_old` with `hyp_Eb` = 30 MeV, keeps 24% of the Σ^0 it produces). The corrected GENIE and NuWro agree on production to 0.96; the remaining factor 4.4 in surviving Σ^0 is FSI. The MicroBooNE Λ paper (arXiv:2212.07888) notes the same asymmetry: FSI *adds* Λ, so there NuWro sits above GENIE. A NuWro control run with ordinary CCQE instead of the hyperon channel reproduces GENIE's CCQE rate to 17%, so NuWro's flux and target normalisation is sound (App. B). The measured Λ cross section, $2.0^{+2.2}_{-1.7} \times 10^{-40}$ cm^2/Ar, brackets both generators. The factor of about two in Σ^0 reduction quoted for NuWro in the second-class-current study is not reproduced by this build; that is open.

2.4 GENIE's hyperon cross section and the correcting weights

With NuWro's cascade switched off, the generators agree on Λ production (NuWro/GENIE 1.07) but not on Σ^0 or Σ^- (0.59 each), with the same energy dependence in all three (Fig. 1, centre). Both use the same Cabibbo SU(3) form factors. An independent calculation from explicit Dirac traces, validated on Llewellyn Smith CCQE, gives $\Sigma^0/\Lambda = 0.19$ on a free proton, in agreement with NuWro (0.21 on argon), against GENIE's 0.45. GENIE's `PaisQELLambdaPXSec` reproduces the trace when the final baryon has the nucleon mass and diverges as the hyperon mass is raised: its hyperon-mass terms are wrong. Relative to the trace, GENIE's Λ is $\times 0.80$ and its Σ^0 and Σ^- $\times 1.85$. The derivation and the checks are in App. A.

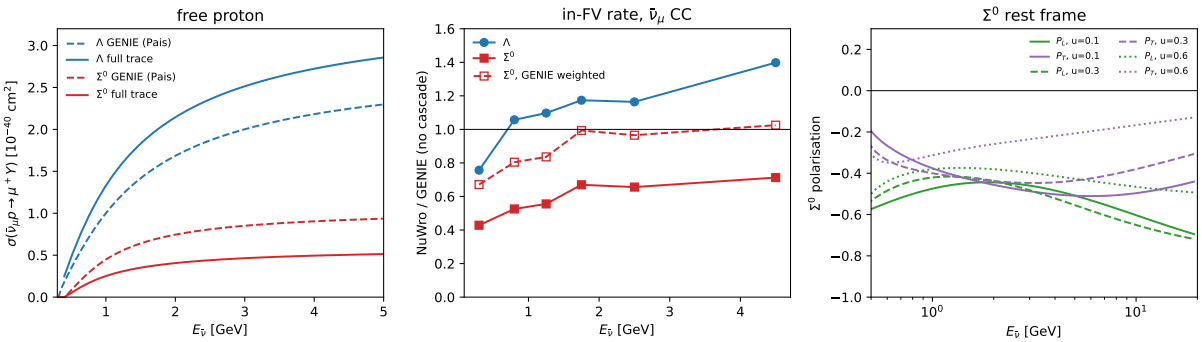


Figure 1: Left: $\bar{\nu}_\mu p \to \mu^+ Y$ on a free proton, GENIE's Pais splines (dashed) and the full trace with the same form factors (solid). Centre: NuWro/GENIE in-FV rate with the NuWro cascade off, flux-weighted, against true E_ν; open squares are GENIE Σ^0 with the cross-section weight. Right: Σ^0 polarisation in its rest frame against E_ν at three values of $u = (Q^2 - Q^2_{\min})/(Q^2_{\max} - Q^2_{\min})$.

Cross-section weight. $w(E, u) = [d\sigma/dQ^2]_{\text{trace}}/[d\sigma/dQ^2]_{\text{Pais}}$ on a free nucleon, with E the $\bar{\nu}_\mu$ energy in the struck-nucleon frame and $u = (Q^2 - Q^2_{\min})/(Q^2_{\max} - Q^2_{\min})$ (`HypXsecWeight.h`, `hyp_xsec_weights.root`). For a two-body reaction E and Q^2 fix the final state up to a flat azimuth, so the weight corrects the kinematics as well as the rate; GENIE's phase space is populated everywhere except a corner holding $\leq 0.3\%$ of the corrected cross section (weight capped at 5). It costs no statistics. On the generator files:

in-FV per data POT	GENIE	GENIE weighted	NuWro (no cascade)	NuWro/GENIE weighted
Λ	209.5	263.0	223.7	0.85
Σ^0	81.7	54.4	47.9	0.88
Σ^-	194.9	129.3	115.0	0.89

The common 0.85–0.89 is a nuclear and flux difference. `SigmaSelection.C` applies the weight to every hyperon-QE event of every GENIE sample (overlay, dirt, detector variations, the QE sample), the struck nucleon taken at rest (1% effect); the hypana chains apply it by event identifier. NuWro is not weighted.

Σ^0 polarisation. The same traces give the Σ^0 polarisation, $P_L \approx P_T \approx -0.4$ to -0.5 (Fig. 1, right). Both generator chains hand Geant4 an unpolarised Σ^0, which decays isotropically. `PolWeights.C` weights each signal event by $w = 1 - \alpha(\vec{P}_\Sigma \cdot \hat{n})(\hat{n} \cdot \hat{q})$ ($\hat{n}$ the Λ direction in the Σ^0 frame, $\hat{q}$ the nucleon direction in the Λ frame, $\alpha = 0.747$), with the decay directions from the hypana ntuple of the same production. $\langle w \rangle = 0.997$, rms 0.15; the selected yields move by 0.2–1.4%.

What the weights cannot correct. The hyperon's nuclear potential and binding near threshold, which shift its kinetic energy, and hyperon FSI, which changes the topology; GENIE has neither. The polarisation weight is needed even with a fixed GENIE.

2.5 FHC

The wrong-sign $\bar{\nu}_\mu$ in FHC gives 0.90× the RHC Σ^0 yield per POT (GENIE, corrected: 49.2 against 54.4 per 1.1526×10^{21} POT) with a similar spectrum ($\langle E_\nu \rangle$ 1.56 against 1.48 GeV). With the 1.1048×10^{21} POT of all FHC runs, including Run 5, that is 47 expected in-FV Σ^0, 87% of RHC. The selection does not use the muon charge, so FHC's ν_μ-dominated background per POT will be larger. The PeLEE FHC overlay (Runs 1, 2, 4c, 4d, 5) and beam-off samples exist; the FHC GENIE signal sample is being regenerated (Sec. 6). FHC enters as a second bin of a combined limit.

3 The PeLEE selection

3.1 Candidates and cut chain

- **Muon**: longest track with `trk_score` > 0.8, start within 20 cm of the vertex, length > 10 cm, LLR PID > 0.7.
- **Proton and pion**: each non-muon track is scored for both roles and the best ordered pair is taken (`PidAssign.h`); pion-candidate purity 42.5% against a ceiling of 51.4% (the fraction of selected signal with a reconstructed true π^-). The candidates are formed when the tracks exist but are not required: requiring them removes more signal than background. Events without a pair carry sentinels and the BDT input `HasPair`.
- **Photon**: the non-track PFPs closest to the vertex; the closest is the Σ^0 decay photon, 70 MeV on average in signal against 74.5 expected.

Stages: `AllEvents`, `FiducialV`, `CosmicVeto` (`topological_score` > 0.20), `NShowers` ($0 < n_{\mathrm{shr}} < 4$), `MuonCandidate`, `NTracksCut` ($n_{\mathrm{trk}} < 5$), `BDTG`.

3.2 BDT

Gradient BDT (600 trees, depth 3) over 57 inputs: candidate PID and geometry, Λ and Σ^0 masses, Λ and photon vertex pointing, Λ decay distance, a π^0 di-photon mass, the Molière angle and the second-shower-veto block of the single-photon analyses (App. D). Two-fold cross-validation by event parity: each fold is applied only to events it did not train on, so every BDT number is out of sample. Each fold trains on 95% of its events; the background tree includes EXT; the signal is the NuWro signal plus the GENIE QE signal scaled to the same summed weight; every input is mapped into its bulk range (`BDTClamp.h`), because the ntuples carry $\pm$FLT_MAX, 10^5 and INT_MAX sentinels that defeat TMVA's evenly spaced cut grid. AUC 0.9086 on NuWro, 0.920 on GENIE. The measurements behind each choice are in App. C.

3.3 Results

Table 4: PeLEE chain, with all weights. Raw signal and S are NuWro; GENIE in Table 5. B = NuWro non-signal hyperons + overlay + dirt + EXT.

stage	raw signal	S	other hyperons	overlay	dirt	EXT	B
AllEvents	21521	12.55	41.33	303821	282305	2406133	2992301
FiducialV	17640	10.31	31.46	90642	22496	148091	261260
CosmicVeto	15619	9.11	26.41	62204	7720	22928	92879
NShowers	12112	7.03	12.00	28995	2906	11942	43856
MuonCandidate	11107	6.46	9.33	20348	1777	6844	28978
NTracksCut	10931	6.36	9.27	18053	1756	6730	26549
BDTG > 0.779	**3686**	**2.117**	**1.08**	**331.4**	**1.93**	**10.50**	**344.9**

Efficiency 16.9% (NuWro, weighted), purity 0.61%, $S/\sqrt{S+B} = 0.114$. Figure 2 shows the score. 0.779 is the peak of $S/\sqrt{S+B}$; there a 2% background systematic is ± 6.8 events against 2.1 signal, so the cut should follow the systematic budget: the expected limit prefers 0.735 at $f = 0$, 0.885 at 10%, 0.910 at 15% (Sec. 5). The surviving background is dominated by resonant and DIS events with a π^0 (App. D).

3.4 GENIE signal

The GENIE QE sample and the overlay's own GENIE Σ^0 events run through the full selection and are counted apart from every sample. The BDT trains on the QE sample's signal with the

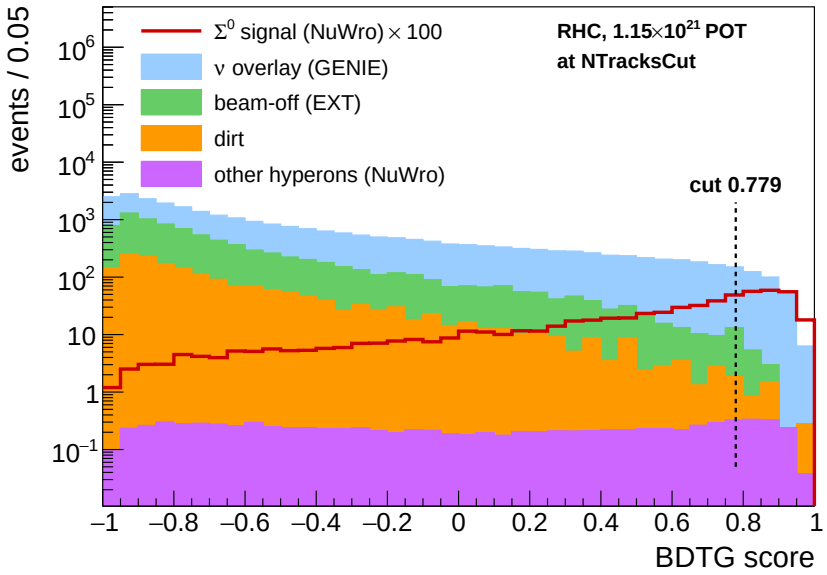


Figure 2: Cross-validated BDTG score at `NTracksCut`, background stacked by sample and the NuWro signal scaled by 100. The dashed line is the operating point.

same parity split, so its yield is out of sample; the overlay's Σ^0 are never trained on. The two agree to 1.5% at every cut.

Table 5: GENIE signal through the PeLEE chain, with the cross-section weight (both) and the polarisation weight (QE sample).

	QE sample			overlay Σ^0		
stage	raw	S_{GENIE}	eff. [%]	raw	S	GENIE/NuWro
AllEvents (true FV)	84089	49.27	100	1038	48.59	3.93
MuonCandidate	43760	25.55	51.9	553	25.80	3.96
NTracksCut	43183	25.21	51.2	544	25.41	3.96
BDTG	15304	8.86 ± 0.07	18.0	187	8.73 ± 0.67	4.19

The efficiencies are 18.0% (GENIE) and 16.9% (NuWro), so the limit on the number of signal events is model-independent to 6%; what depends on the model is the rate it is compared with.

3.5 Detector variations

The Run 3 RHC detector-variation set re-simulates the same events under nine variations; `DetVar.C` compares each with the CV on shared events (old-flux samples, so only ratios are used).

variation	S var/CV	B var/CV
LY attenuation	1.000 ± 0.000	1.005 ± 0.005
LY down	0.975 ± 0.025	0.991 ± 0.006
LY Rayleigh	0.975 ± 0.039	0.984 ± 0.009
recombination	1.121 ± 0.072	0.926 ± 0.030
SCE	0.900 ± 0.108	0.972 ± 0.035
WireMod θ_{XZ}	0.992 ± 0.078	0.970 ± 0.025
WireMod θ_{YZ}	0.906 ± 0.087	0.976 ± 0.027
WireMod X	0.956 ± 0.078	0.972 ± 0.027
WireMod YZ	0.945 ± 0.081	0.985 ± 0.025
quadrature (MC noise alone)	19.9% (21.3%)	9.6% (7.0%)

At 0.779, with all weights. The signal is not resolved (its quadrature sum equals the MC noise). The background is: recombination lowers it by $7.4 \pm 3.0\%$ (2.5σ), and the sum exceeds the noise by about 6.6% in quadrature. The CV keeps 51 raw signal and 1 174 raw background events, so the background detector systematic is about 7–10% and the signal one is not measured.

4 The Λ-style selection

The MicroBooNE Λ CCQE analysis (Thorpe & Nowak, Hyperon Technote v2) selects $\bar{\nu}_\mu$ CC Λ with 91% purity, where the earlier hypana BDT chain for Σ^0 reached 2.8% at the same efficiency. This chain follows it:

- **Every ordered track pair is scored**; the pair selector's only topological input is the separation of the two track starts, which does not penalise a Λ that flew. Pair precuts are a preference, not a veto.
- **The decay vertex is fitted** (V fit), giving a decay length and the pointing angle α between the pair momentum and the vertex-to-decay direction.
- **The island test** (Fig. 3): on each plane a wire × time grid of the raw activity is clustered from seeds at the muon, proton and pion starts, and the proton and pion clusters must merge while the muon stays separate. Connectivity 2 wires × 20 ticks passes 80% of signal with 2–3 cm Λ flight. It needs the raw wire windows, hence hypana.

4.1 Operating point and results

Λ mass within ±20 MeV, decay length > 0.5 cm, $\cos\alpha > 0.95$, photon start > 2 cm from the vertex, island test passed on ≥ 2 planes, and a proton-like proton candidate (`llrP` < 0, Sec. 4.4). With the NuWro signal $S = 0.126 \pm 0.009$ (219 raw). NuWro's own non-signal events (Σ^0 that FSI turned into something else) add 0.020, and EXT 0.34 ± 0.15 (Sec. 4.2), so $B = 2.48$. 98.6% of the surviving signal has true Λ flight > 2 cm. The signal efficiency is 1.0% for both generators.

4.2 Beam-off background

The run3b `ext_unbiased` files were taken through reco2 and HyperonProduction: 16 570 events, 0.67% of the data's cosmic statistics, too little to follow to the operating point (none of 76 Muon-Candidate events reaches the pointing cut). `ExtFactor.C` factorises: EXT at MuonCandidate times the pass fraction of the overlay's cosmic-dominated candidates as far as the pointing cut, and the neutrino-background rate (3.3%) for the island and gap steps. EXT is 0.62 ± 0.27 events before the PID cut and, scaled by the MC background's PID pass fraction (0.54), 0.34 ± 0.15 **at the operating point**. Without the neutrino-rate assumption for the island step the bound is

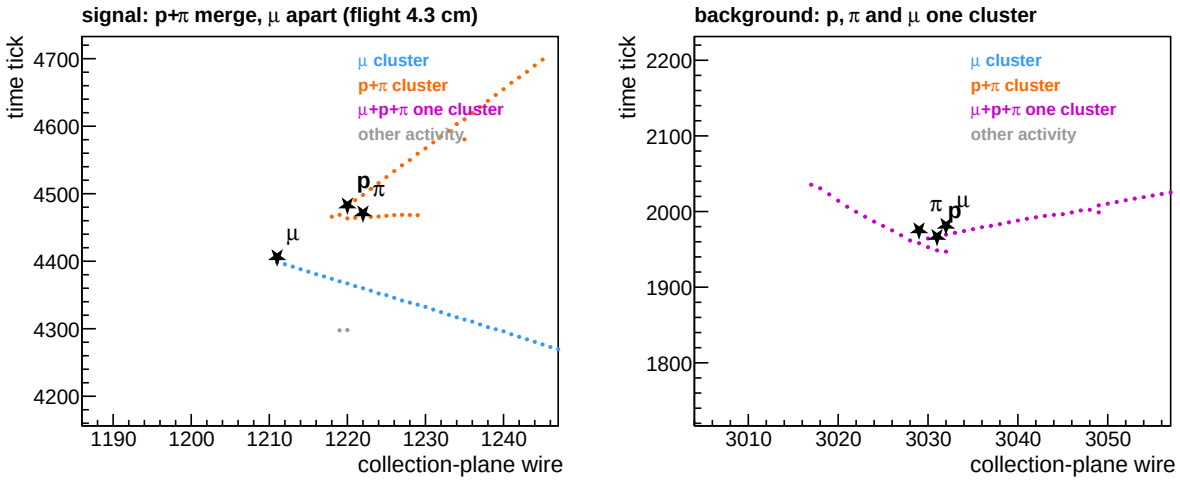


Figure 3: The island test on the collection plane. Left: a GENIE signal event with 4.3 cm true Λ flight; the proton and pion seeds share a cluster (orange) and the muon has its own (blue). Right: a background event at the same stage, whose proton, pion and muon seeds land in one cluster.

Table 6: Λ-style cut flow. S is the GENIE Σ^0 QE sample with the cross-section, polarisation and PPFX weights (the first two stages precede the island cache). B is the hypana overlay, with the cross-section weight at the last stage. Errors are MC statistics.

stage	raw signal	raw background	S (GENIE)	B
AllEvents	–	1840942	–	301031
SliceID	–	776242	–	126931
MuonCandidate	47033	123000	27.41	20113
PairCandidate	13226	51238	7.75	8378
Λ mass ±20 MeV	7527	16391	4.39	2680
pointing ($\cos\alpha$, decay length)	2065	706	1.21	115.4
photon gap > 2 cm	1971	549	1.16	89.8
island, ≥ 2 planes	956	23	0.564 ± 0.018	3.92 ± 0.82
proton PID, `llrP` < 0	**875**	**12**	$\mathbf{0.516 \pm 0.018}$	$\mathbf{2.12 \pm 0.62}$

< 6.2 at 90% CL; that, and the PID scaling, are the open part of this estimate. The real cosmic data agree with the factorisation wherever they still have events (8 of 76 EXT candidates have a pair, against 0.135 predicted from the cosmic MC and 0.42 from the neutrino rate). Details are in App. F.

4.3 What the remaining background is

By the true identity of the candidates, before and after the proton PID cut:

category	raw before	raw after	B after (weighted)
real $\Lambda \to p\pi^-$ (GENIE $\bar{\nu}_\mu$ CC QE Λ)	4	4	0.81
real p and π^- from Δ/N* or DIS, roles right	4	4	0.65
real p and π^-, roles swapped	3	0	–
pp, $p\pi^+$ (the pion candidate is not a π^-)	4	3	0.49
$\pi^+\pi^+$, $\pi^+\pi^-$, $\mu\pi^+$ (not a proton)	4	1	0.16
two showers reconstructed as the pair	4	0	–
total	23	12	2.12

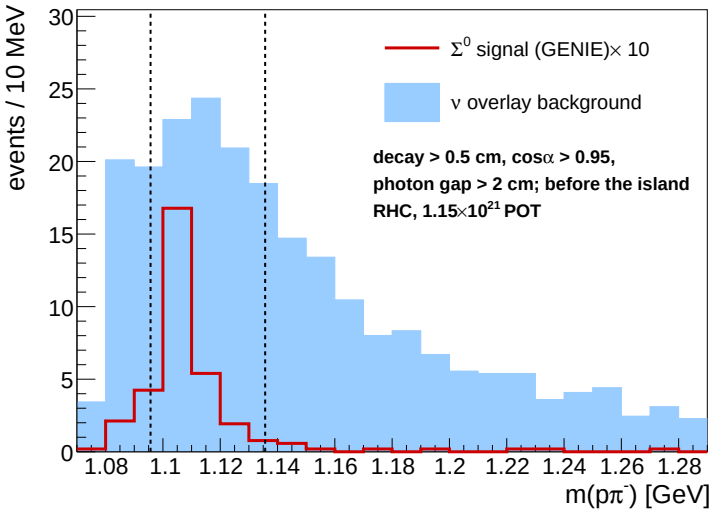


Figure 4: $p\pi^-$ mass after the pointing cut and the photon gap, before the mass window (dashed) and the island test. The pion momentum is taken from range under the muon hypothesis, 20% low for a pion but with a narrower peak than any correction. The background is largely genuine $p\pi$ from Δ decay.

The real Λ is now 38% of the MC background. Its pair is a genuine Λ decay, so only the photon tells it apart: in these events the chain's "photon" is a Michel positron from the μ^+ decay (starting 0.6 cm from the muon's end), two δ-ray electrons on the muon track, and an overlaid cosmic, at 66–452 cm from the vertex.

4.4 Selection optimisation

The figure of merit is the expected limit; at $B \approx 4.6$ it is set by Poisson counting, so signal efficiency matters as much as rejection. The background rests on 23 raw events, so every candidate cut is checked at the photon-gap stage (549 raw) and by parity cross-validation (`IslandOptimise.py`).

- **Proton-candidate PID (`llrP` < 0): adopted.** It keeps 91.5% of signal for both generators and removes about half the background. The expected limit improves by 12–13% on the median and on the mean over outcomes, and on both parity halves; its threshold is the natural midpoint, not tuned.

- **After it, the real Λ is 38% of the background.** Its pair is genuine, so only the photon separates it. A maximum photon distance removes it but loses the 16% of signal photons that convert beyond 55 cm, and does not improve the limit. Shower shape (hits per plane, length, opening angle, Molière angle, dE/dx) does not separate the Michel and δ-ray electrons from the soft Σ^0 photon. Photon pointing to the vertex rejects the real Λ but the 45 MeV photon is too poorly reconstructed for a useful cut. The Σ^0 mass window costs more signal than it removes. **A muon-track photon veto** (reject the event if its photon starts within 5 cm of the muon axis and more than 20 cm along it, up to 10 cm past its end: δ rays and Michel electrons) removes 2 of the 4 real-Λ events for 6% of signal, but leaves the mean expected limit unchanged (9.32 against 9.38 GENIE); it is not used. Without the 20 cm the veto would catch 23% of true Σ^0 photons, those converting near the vertex in the muon's direction (App. F).

- **Loosening any single cut** (mass window, pointing, decay length, photon gap, island planes, PID threshold) recovers signal only with proportionally more background and does

not improve the limit. A full scan finds 20% on the full sample but about 6% out of sample; the background statistics do not support a finer optimisation.

- **Events without a reconstructed shower** (the soft Σ^0 photon missed, 20% of the signal at that stage) were taken through the chain as a second bin: $S = 0.213$ GENIE on $B = 5.64$ (EXT 1.53 ± 0.88 measured with the hypana EXT sample), purity 3.7%. Merged with the main bin they worsen the limit; as a separate bin they would gain 7–8%, which is not worth a second category. They are not used (App. F).

5 Expected sensitivity

5.1 Method

`Sensitivity.C` gives the expected 90% CL upper limit on μ, the Σ^0 rate in units of a generator's prediction, for both chains under both generators. Each chain is a single-bin counting experiment, $n \sim \text{Pois}(\mu S e + b)$:

- b is lognormal with mean B and fractional width $\sqrt{r_{\text{MC}}^2 + f^2}$, r_{MC} the MC statistics of B and f the background systematic, which is scanned;
- e is lognormal with mean 1 and width $\sqrt{g^2 + r_S^2}$, $g = 20\%$ a placeholder for flux, efficiency and POT;
- Bayesian limit, flat prior in $\mu \geq 0$, b and e marginalised (hybrid CL_s is 4% tighter; both reproduce the PDG table for known b);
- expected limit at the median n of Pois(B), band from the 16% and 84% quantiles; Z_A from Cowan et al. (arXiv:1007.1727).

$\mu_{\text{UL}} = n_{\text{UL}}(B, \sigma_b)/S$: the generator only rescales the limit, and the cut that minimises it does not depend on the signal normalisation.

5.2 Results

Table 7: Median expected 90% CL limit on μ, RHC, $g = 20\%$, with all weights. "Re-optimised" scans the BDT cut from 0.50 to 0.95 for each f (cut in brackets); its GENIE value is the NuWro value divided by the GENIE/NuWro ratio at 0.779. Λ-style with the proton PID cut.

f	PeLEE, BDTG > 0.779		PeLEE, re-optimised		Λ-style	
	NuWro [16–84%]	GENIE	NuWro (cut)	GENIE	NuWro [16–84%]	GENIE
0%	17.3 [12.1, 25.1]	4.13	17.2 (0.735)	4.11	33.2 [25.9, 51.5]	8.04
5%	22.5	5.37	21.9 (0.825)	5.23	33.2	8.05
10%	32.8	7.83	26.0 (0.885)	6.20	33.3	8.07
15%	43.9 [37.1, 51.5]	10.48	29.5 (0.910)	7.04	33.4 [26.0, 51.9]	8.10
20%	54.7	13.06	33.0 (0.925)	7.88	33.6	8.15
30%	74.4 [66.6, 82.6]	17.76	36.0 (0.945)	8.59	34.1 [26.4, 52.8]	8.27

1. **The best expected limit**, about 4.1× GENIE and 17× NuWro, is about 210 in-FV Σ^0 events in 1.15×10^{21} POT, against 49 predicted by GENIE and 13 by NuWro. In events the two models agree to 6% (217 and 204 at $f = 0$), because the efficiencies do.

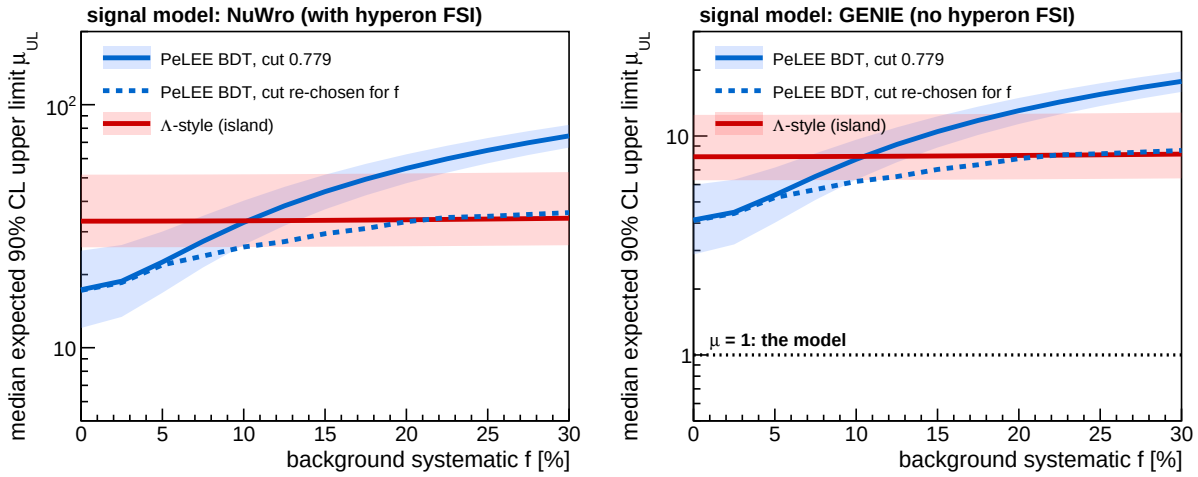


Figure 5: Median expected 90% CL limit on μ against the background systematic, under NuWro (left) and GENIE (right). Bands are the 16–84% range; the dotted line on the right is the GENIE prediction itself.

2. **The PeLEE chain gives the better limit below $f \approx 21\%$, the Λ-style chain above it**, under both generators. With its cut set for the systematic the PeLEE chain is 1.9× better at $f = 0$, 1.3× at 10% and 1.1× at 15%; at 30% the Λ-style chain leads, 34.1 against 36.0 (NuWro) and 8.27 against 8.59 (GENIE). The Λ-style signal is 17× lower, but at $B \approx 2.5$ its limit is Poisson-limited and nearly flat in f: removing all background uncertainty moves it by 2%.

3. **The background systematic drives the PeLEE result.** Its limit grows 1.7× from $f = 0$ to 15% with the cut re-optimised and 2.5× at 0.779; at $f = 10\%$ moving the cut to 0.885 gains 21%.

4. **Nothing missing from the Λ-style chain changes its limit much** ($f = 15\%$, NuWro): without EXT 34.6 against 33.4; with dirt at its bound (8.7% of B_{MC}) 32.9; with $g = 0$ 30.6; with B known exactly 32.6. The median n is an integer (2 here), so small changes in B move the median limit in steps, either way. With the PeLEE systematic above $\sim 20\%$ the Λ-style chain would carry the result, which makes hypana data and dirt worth producing.

6 Open items and plan

1. **Background systematics for the PeLEE chain (priority).** Flux (`weightsPPFX`), cross section (`weightsGenie`) and reinteraction (`weightsReint`) multisims at the single bin, with the detector variations (7–10%). Then a data constraint from the low-score BDT region, the π^0 normalisation being the obvious target. Choose the BDT cut from the resulting budget and re-measure the detector variations there.

2. **Fixed GENIE signal samples.** `genie::HypQELTracePXSec` (App. A.1) is validated; RHC (3050 chunks) and then FHC (1200 chunks) are being regenerated, at the lowest batch priority, on all nodes. They then go through the overlay chain to PeLEE and hypana ntuples. With them the cross-section weight is dropped for the signal samples; the polarisation weight stays, and the overlay and dirt keep the cross-section weight. The GENIE problem should be reported to the GENIE authors: it moves GENIE's Λ prediction by ×1.26, the model the MicroBooNE Λ measurement compared with.

3. **FHC** (Sec. 2.5): the FHC background through the PeLEE selection, the FHC signal efficiency from the regenerated sample, possibly a separate FHC cut, and a two-bin combined limit with correlated signal normalisation.

4. **Λ-style chain.** Retune the island on the QE sample (it was tuned on the duplicated hyperon sample); redo the Λ cross-check against the published GENIE prediction (App. F). Its EXT estimate (the island step and the PID scaling) is the open part of its background. If the PeLEE background systematic comes out above $\sim 20\%$, this chain carries the result, and data and dirt through reco2 and hypana become necessary rather than a cross-check.

5. **Signal model.** Hyperon FSI is the factor four between the generators, and the NuWro build reproduces neither the published FSI reduction nor, so far, a reason for the difference; the limit is quoted under both models.

6. **Closure and unblinding.** A Poisson-throw fake-data closure test (App. H) with the GENIE sample as the alternative model; then the observed limit from the RHC beam-on data (`custom_pelee_ntuples`), and the absolute cross-section limit from the integrated $\bar{\nu}_\mu$ flux and the argon target count.

A GENIE's hyperon cross section in detail

Generators with the cascade off. NuWro was rerun with no cascade and no jailing (`kaskada_on=0`, `hyp_Eb=0`), one hyperon channel at a time, on the configuration of the signal production with the flux weight; GENIE is taken from the generator files of the QE production with the relevant hyperon kept (`genie_to_hepmc.C`). In-FV $\bar{\nu}_\mu$ CC events with a primary hyperon per data POT (`GenRate.C`):

	NuWro	GENIE	NuWro/GENIE
Λ	223.7 ± 2.9	209.5 ± 1.4	1.07
Σ^0	47.9 ± 0.6	81.7 ± 0.6	0.59
Σ^-	115 ± 2	194.9 ± 1.4	0.59

Σ^-/Σ^0 is 2.40 in NuWro and 2.38 in GENIE: the isospin factor 2 times argon's N/Z (2.44). The NuWro/GENIE ratio rises the same way with E_ν for Λ and Σ^0; only the normalisation differs, so neither the flux nor the nucleus is the cause.

Same couplings. Both generators use $f_1^\Lambda = -\sqrt{3/2}\,F_1^p$, $g_1^\Lambda = -(1+2x)g_A/\sqrt{6}$, $f_1^{\Sigma^0} = -(F_1^p + 2F_1^n)/\sqrt{2}$, $g_1^{\Sigma^0} = (1-2x)g_A/\sqrt{2}$, $\Sigma^- \sqrt{2} \times \Sigma^0$, $x = F/(F+D) = 0.365$ and $M_A = 0.99$–1.0 GeV (GENIE `LwlynSmithFF::Strange*`, NuWro `ff.cc`). NuWro contracts the full tensors (`Singh_Model`); GENIE uses a closed form.

Independent calculation. `HypXsecWeights.py` evaluates $L_{\mu\nu}H^{\mu\nu}$ from explicit Dirac matrices, L from $\bar{v}(k_1)\gamma^\mu(1-\gamma_5)v(k_2)$ and H for $F_1\gamma^\mu + i\xi F_2\sigma^{\mu\nu}q_\nu/2M + F_A\gamma^\mu\gamma_5$. On CCQE it gives 1.06 (ν) and 0.41 ($\bar{\nu}_\mu$) $\times 10^{-38}$ cm^2 at 1 GeV, the Llewellyn Smith values. Free proton, 10^{-40} cm^2:

	trace			GENIE spline		
E_ν	Λ	Σ^0	Σ^0/Λ	Λ	Σ^0	Σ^0/Λ
1 GeV	1.33	0.25	0.19	1.02	0.46	0.45
2 GeV	2.15	0.41	0.19	1.71	0.76	0.44

Where GENIE goes wrong. A transcription of `PaisQELLambdaPXSec::XSec` (identical in v3_00_04_ub4 and v3_00_06) reproduces GENIE's Λ spline (1.05 against 1.02 at 1 GeV; its Σ^0 is 15% under GENIE's, which has $G_E^n \neq 0$). With the final-state mass set to the nucleon's it agrees with the trace to 0.5%. As the mass is raised to the hyperon's, the trace cross section falls, as phase space requires, while the Pais Σ^0 cross section rises ($0.49 \to 0.60$ at 2 GeV, against $0.50 \to 0.41$): the error is in the $M_Y - M_N$ terms.

Polarisation. The spin projector $(1 + \gamma_5 \not{s})/2$ on the hyperon, in its rest frame reached from the struck-nucleon frame, gives P_L, P_T (in the production plane, towards the $\bar{\nu}_\mu$) and $P_N = 0$. The conventions are checked against explicit helicity spinors and on a point-like $V - A$ current ($|\vec{P}| = 1$ at every angle, $P_L \to -0.90$ for a fast hyperon; helicity -1 is forbidden only in the exactly backward corner). The M1 decay passes the polarisation on as $\vec{P}_\Lambda = -(\vec{P}_\Sigma \cdot \hat{n})\hat{n}$. The rebuilt decay directions give 75.4 ± 10 MeV/c for the Λ in the Σ^0 frame (74.5 expected) and 108 ± 26 MeV/c for the nucleon in the Λ frame (100.6).

NuWro's cascade on Λ. Of primary Λ, 64% leave the nucleus as Λ, 32% are jailed, 4% convert to Σ; of primary Σ^0 24% survive.

A.1 The fixed GENIE

`genie::HypQELTracePXSec` (directory `genie_gen/hypfix` of the production area) replaces `PaisQELLambdaPXSec` for the `QEL-CC-LAMBDA` generator: same channels, form factors, Cabibbo angle, integrator and nuclear scaling, with $L_{\mu\nu}H^{\mu\nu}$ evaluated from Dirac matrices. GENIE is not rebuilt: the class lives in its own library, autoloaded by ROOT through its rootmap, and XML overrides on `GXMLPATH` (`ModelConfiguration.xml`, `master_config.xml`) point the generator at it. 387 new splines cover the 44 nuclei of the production spline file for $\bar{\nu}_e$, $\bar{\nu}_\mu$, $\bar{\nu}_\tau$ (250 knots to 1000 GeV), merged into a copy of `gxspl-FNALsmall.xml`. Validation:

- the C++ contraction reproduces the trace exactly (1 GeV: Λ 1.330, Σ^0 0.252, Σ^- 0.499 $\times 10^{-40}$ cm^2);
- GENIE splines, new/Pais: Λ 1.268/1.218/1.189 at 1/2/5 GeV against 1.269/1.216/1.185 expected; Σ^0, Σ^- 0.665–0.698 against 0.656–0.687 ($G_E^n \neq 0$ adds $\sim 2\%$);
- a 360-event FHC `lar` generation: primary Λ $59.2 \pm 2.6\%$, Σ^- $30.1 \pm 2.4\%$, Σ^0 $10.7 \pm 1.6\%$, against 42.8/40.4/16.7% with Pais and about 58/30/12% expected. Its POT per event is 1.8σ below the +9% expected from the lower total cross section; to be checked on the production.

At the same chunk count the regenerated samples hold about 0.7× the Σ^0 statistics of the old ones.

B Samples: history and validation

NuWro run3b production. 2500 jobs of 100 NuWro Σ^0 events overlaid on run3b `ext_unbiased` data through gen→reco2, with hypana and PeLEE ntuples; it replaced the Run 1-EXT sample in October 2026 (3.3× the raw statistics, cosmics from a period the background overlay covers). POT agrees between the two formats to four digits; no EXT event is used twice.

Background sample corrections. The file first used as "Run 1 dirt" was FHC dirt (its E_ν spectra match FHC, $\chi^2/\mathrm{ndf} \approx 1$, not RHC, 90–660); it was replaced by RHC dirt, which is on the old flux (`ppfx_cv` profile) and takes the NuWro map. The Run 3 overlay (55% of the statistics) was missing. EXT is scaled by beam-on triggers (23 534 937 per 1.10823×10^{21} POT); data files have empty `backtracked_*` vectors, and 1 655 repeated events are dropped. The overlay's own 1 038 raw in-FV GENIE Σ^0 events are kept out of B and counted as the GENIE signal prediction.

GENIE Σ^0 QE sample. GENIE `LambdaCCQE`, tune G18_10a_02_11a, official dk2nu RHC flux, primary Σ^0 selected at generation, same overlay chain; 2506 jobs, 1.23005×10^{24} POT. Before the cross-section correction its in-FV rate, 81.85 ± 0.28, agreed with the overlay's own Σ^0 (83.39 ± 3.69, 0.98 ± 0.04) and in shape (Fig. 6). The overlay's non-QE Σ^0 (0.8%) come with a kaon, which the signal definition vetoes, so the QE sample is the complete GENIE prediction. The previous GENIE hyperon sample (`sig_genie.list`) held 15 341 events but only 754 distinct interactions, each overlaid about 20 times, at 1.32× the overlay's rate; everything that used it was redone.

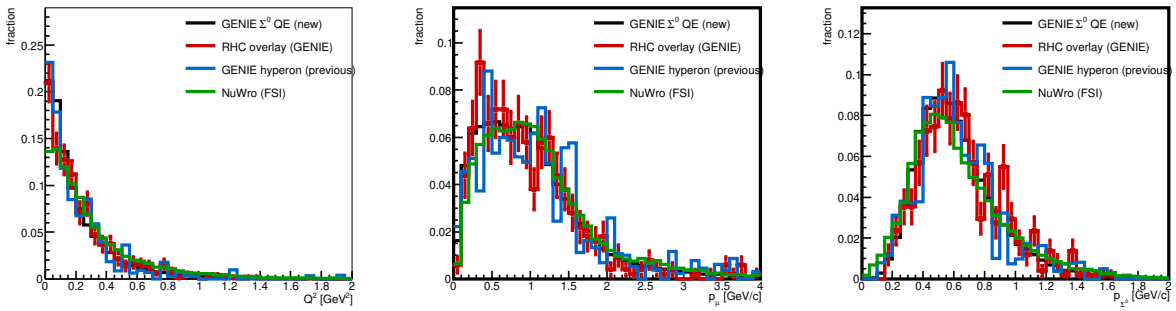


Figure 6: Truth Q^2, muon momentum and Σ^0 momentum of in-FV $\bar{\nu}_\mu$ CC Σ^0, area-normalised: the GENIE Σ^0 QE sample, the plain RHC overlay, the previous GENIE hyperon sample and NuWro (before the cross-section correction).

The rate comparison before the correction. Before GENIE's cross section was corrected, the 6.5× gap between the GENIE overlay (83.3) and NuWro (12.8) was decomposed as FSI (4.19×), flux weight (1.48×) and production (1.05×). A NuWro control run with `dyn_hyp_cc=0` `dyn_qel_cc=1` gives 7.539×10^{15} POT per CCQE event against the overlay's 8.835×10^{15} on the same footing, so NuWro's generic rate is within 17% of GENIE's. With the correction the gap is 4.4× and production agrees to 0.96.

C PeLEE BDT development

Out of sample on NuWro (signal statistics limit the BDT, so every gain came from more or better-used events):

training	AUC	B/B_0 at signal efficiency			peak $S/\sqrt{S+B}$	
		50%	40%	20%	value	cut
no EXT, half fold	0.8894	0.0412	0.0253	0.0067	0.0993	0.690
EXT, half fold (3 seeds)	0.8956–0.8966	0.0378–0.0385	0.0228–0.0236	0.0059–0.0062	0.1037–0.1049	
EXT, 95% of fold	0.9000	0.0358	0.0215	0.0061	0.1077	0.759
clamped inputs	0.9057	0.0336	0.0203	0.0063	0.1102	
clamped + GENIE signal (adopted)	**0.9086**	**0.0317**	**0.0186**	**0.0055**	**0.1146**	**0.779**
clamped + GENIE, seed 1	0.9093	0.0312	0.0186	0.0055	0.1147	

- **EXT in training**: +5% figure of merit, almost all EXT rejection.

- 95% **of each fold**: TMVA's default trains each fold on half its events (25% of all); +4%, every full-fold seed beats every half-fold seed.
- **Input clamp**: one ±FLT_MAX value leaves a variable unusable in a node because TMVA's cut grid is evenly spaced; with bagging almost every root node contained one of five events with infinite masses. The five pair distances carry a 10^5 sentinel in 21–59% of events, `ShrSubClusters` INT_MAX in 29%.
- **GENIE signal in training**: more distinct signal helps even judged on NuWro; −5% background at 40–50% efficiency.

The BDT trains on unweighted GENIE events; with the hyperon weights its scores are unchanged and its yields correctly weighted, so it was not retrained.

D What limits the PeLEE purity

Measured on earlier versions of the chain; the fractions carry over.

- **The surviving background is π^0 events**: resonant 52%, DIS 24%, QE 14%, MEC 7%, cosmic 3%. A π^0 supplies the fake photon and its hadrons the fake proton and pion.
- **The candidate assignment is the weak link**: in background the muon candidate is right 81% of the time, the proton 58%, the pion 31%, the photon 29%.
- **The signal photon is soft**: $E_\gamma = (m^2_{\Sigma^0} - m^2_\Lambda)/2m_{\Sigma^0} = 74.5$ MeV, against 140 MeV for background photons. Soft-photon cuts help at preselection but not on top of the BDT, which already uses the energy; a π^0 mass veto rejects only 9% because it needs two showers; the Molière angle adds 1%.
- **Methods from the single-photon analyses**: the second-shower-veto variables are BDT inputs; one BDT per background class and a `HasPair` split were tried and do not help; the conversion distance is the Λ-style photon gap; the π^0 normalisation (GENIE over-predicts NCπ^0 by 25%) belongs in the covariance and a sideband constraint.

E Reconstruction study on HyperonProduction ntuples

The PeLEE truth record never contains the $\Sigma^0 \to \Lambda\gamma \to p\pi^-\gamma$ chain (zero of 17 721 signal events have a backtracked π^-). Hypana writes the full chain and per-PFP truth; `HypInput.h` maps it onto the PeLEE names. Three format traps: no implicit neutrino-slice requirement (an explicit `flash_match_nu_slice_ID` ≥ 0 stage brings the chains to 6–8% agreement), `trk_energy_proton` is the proton momentum (with the conversion the Λ mass width falls from 23.5 to 7.8 MeV), and `pfp_true_origin` is the analyser's own enum (1 neutrino primary, 2 Λ daughter, 3 other, 5 Σ^0 daughter, −1 unmatched).

With both the reconstructed and true vertex in the FV and $\Lambda \to p\pi^-$ in truth (3704 signal events), the four particles are reconstructed as: μ^+ 91.8%, p 60.9%, π^- 47.3%, γ 62.4%; all four together 18.1%, the ceiling for any selection requiring all four. Track/shower: the pion is a track 54% of the time, and the photon a track 15%, which feeds the photon contamination of the pion slot. The Λ flight from the primary vertex is not measurable (median true flight 1.85 cm against 5.5 cm vertex resolution, $r = -0.014$), which is why the Λ-style chain fits the decay vertex from the two tracks instead.

F Λ-style chain: development and checks

Island connectivity. The original 3×40 bridged the decay V to the muon for flights below ~ 3 cm:

true flight	1–2 cm	2–3 cm	3–5 cm	5–10 cm
3×40	19%	45%	70%	86%
2×20	43%	80%	85%	91%

On one plane the figure of merit is flat in the connectivity; the gain is from two planes (6.8× background rejection against 3.7×). Tuned on the Run 3 list and scored on the independent extension, 2×20 is 0.004 from the top of a plateau. The two-plane pass rate is 50% for both generators.

Out-of-sample checks. Without the photon gap, the Run 3 list gives $B = 4.15 \pm 1.31$ and the extension 7.6 ± 1.4; every earlier stage agrees per POT to 1–4%, so the low Run 3 value was a fluctuation. The gap halves the extension's background at 96% signal efficiency. A Michel tag removes 87% of true μ^- but costs 22% of signal.

Sidebands (one-plane point): mass sideband +40 to +200 MeV $S/B = 0.2\%$, α sideband 0.9%, island-fail region 1.1%, all RES/DIS-dominated like the signal region; the mass sideband exists only above the peak ($m(p\pi) \geq 1.078$ GeV).

EXT factorisation.

step	EXT raw	EXT P	MC cosmic raw	MC cosmic P	MC all-bkg P
MuonCandidate	76 ($11\,419 \pm 1\,310$)	–	4 220	–	–
pair	8	0.105 ± 0.035	571	0.135 ± 0.005	0.42
Λ mass	3	0.38 ± 0.17	165	0.289 ± 0.019	0.32
pointing	0	0 of 3	7	0.042 ± 0.016	0.043
island + gap	0	–	0 of 7	–	0.0326

Hypana EXT at MuonCandidate is 1.7× the PeLEE beam-off there, because hypana's `topological_score` is the highest of any slice; the MC cosmics use the same definition.

Λ cross-check. Reimplementing the Λ CCQE chain on our files (`LambdaSel.C`) reproduced its efficiency to 0.79×, but the GENIE signal normalisation came out 1.61× the published one, on the duplicated hyperon sample. It has to be redone on the QE sample with the corrected cross section, which moves GENIE's Λ by ×1.26.

Muon-track photon veto. With the attached zone starting at the vertex (−5 cm along the muon), 23% of signal photons count as attached, against 50% of real-Λ and 33% of other background photons; re-picking the photon among the non-attached showers costs 18% of signal. Starting the zone 10, 20 or 30 cm along the muon flags 8.5, 7.5, 7.2% of signal, 4, 3, 3 of 10 real-Λ events and 9.3, 8.5, 8.0% of other background at the photon-gap stage. On top of the PID cut the 20 cm veto gives $S = 0.484$, $B = 1.99$: median limit 9.03 (the median n stays 2 while S drops), mean 9.32 against 9.38.

The shower requirement. The preselection requires $0 < n_{\mathrm{shr}} < 4$ and $n_{\mathrm{trk}} < 5$ (`PreselFlow.C`). Of the in-FV GENIE signal (51.1 events) 84% has a flash-matched neutrino slice, 82% a reconstructed vertex in the FV, 75% passes the topological score, 57% the shower/track requirement and 50% has a muon candidate; the shower requirement removes 7.8 events, almost all with no shower at all. Relaxing the shower maximum, the track maximum or the topological score is worth 2–5% each. The events with no shower (`IslandProcess.C showerMode=1`) as a separate bin, at the PID operating point:

stage	S GENIE	S NuWro	raw bkg	B_{MC}
no shower, $n_{\mathrm{trk}} < 5$	10.70	2.71	213 731	34 949
muon candidate	9.38	2.40	174 614	28 538
pair	3.31	0.867	51 839	8 472
Λ mass	1.84	0.479	17 754	2 902
pointing	0.499	0.125	606	99.9
island ≥ 2 planes	0.244	0.065	25	4.73
proton PID	0.213	0.059	21	4.04

EXT, factorised as for the main bin from the hypana EXT run in the same mode (161 muon candidates against 76 with a shower), is 1.53 ± 0.88, so $B = 5.64$. The bin's background passes the PID cut 84% of the time (main bin 52%): it is mostly genuine $p\pi^-$ pairs, and there is no photon gap to apply. Pooled with the main bin the mean expected limit worsens (9.98 against 9.39 GENIE); as a second bin it would improve by 7% (GENIE) and 8% (NuWro). Not used.

Background photons. The chain's photon in the real-Λ background is never a Σ^0 photon. Of 10 real-Λ events at the photon-gap stage, half have $e^{\pm}$ (Michel positrons and muon δ rays), 30% another photon, 10% a cosmic. In signal the chain picks the true Σ^0 photon 80% of the time (median 115 hits on the collection plane, 54 MeV, 24 cm long); the real-Λ "photons" are similar (75 hits, 29 MeV, 14 cm) and the rest of the background larger (133 hits, 87 MeV, 28 cm).

G The hypana BDT chain

A BDT on the hypana candidates (53 inputs, including the island variables), used for the truth studies of App. E, not for the limit. Trained on the GENIE QE signal with two-fold parity cross-validation, AUC 0.855. With the hyperon weights (matched by event order to the island caches, which carry event identifiers) at its cut 0.554: $S = 11.32$ (GENIE, no PPFX) on $B = 737.4$, purity 1.5%.

H Methods kept for later

A global hypothesis fit (not built): every PFP eligible for every role, all assignments scored with a combined $-2 \ln L$ from the Λ and Σ^0 mass constraints, per-role PID templates, the p–π DCA and priors; perfect assignment given what is reconstructed is worth 1.54× on events with four PFPs. Its central idea, joint assignment, is realised by the pair scoring of both chains. The measured ingredients are in `HypFitParams.h`; the masses must be kept out of any BDT and the fit run on background alone to check for an induced peak.

Fake data by Poisson throw. The PeLEE "data" slot holds the GENIE QE sample scaled by POT: a generator test of the efficiency, not a closure test. The CC1μ1π Poisson-throw scheme (`throw_perrun_w.C`: $n \sim \mathrm{Pois}(w_{\mathrm{cv}}\alpha)$ per MC event, EXT at its scale) applied to all samples gives the statistical closure test; injecting the GENIE signal into a NuWro-based throw gives the model test.

variable	μ^+	p	π^-	γ
`trk_shr_score`	1.00	0.90	0.80	**0.11**
`trk_llrpid`	**0.96**	−0.29	0.14	0.56
`trk_pid_chipr`	180	**16.4**	96	192
`three_plane_dedx`	2.47	**10.3**	7.52	3.30
`rms_deflection`	0.01	0.01	0.01	**0.08**
`moliere_avg`	1.78	6.18	8.95	**13.3**
`trk_length`	**139**	12.8	10.8	14.2

Table 8: Per-role PID templates, medians on truth-matched PFPs. The π^- has no distinctive signature.

Framework port. A `SelectionBase` subclass in `xsec_analyzer` was planned for its systematics machinery; it is not started and no longer on the critical path, since the PeLEE ntuples carry the multisim weights and `Sensitivity.C` provides the limit.

I Change log

2026-10-15 NuWro run3b signal production replaces the Run 1-EXT sample.

2026-10-17 RHC dirt replaces FHC dirt; Run 3 overlay and EXT added; overlay signal removed from B.

2026-10-18 PeLEE BDT retrained with EXT and on 95% of each fold.

2026-10-19 Detector variations; Λ-style chain and its EXT estimate; expected sensitivity.

2026-10-30 GENIE Σ^0 QE sample replaces the duplicated hyperon sample; PeLEE BDT retrained with clamped inputs and GENIE signal (cut 0.759 → 0.779). GENIE's hyperon cross section found wrong; cross-section and polarisation weights applied everywhere (GENIE signal 13.47 → 8.86, limits 1.5× weaker).

2026-10-31 GENIE fixed (`HypQELTracePXSec`); RHC and FHC GENIE signal regeneration started. Λ-style background weighted; background composition; FHC signal estimate; note restructured around the status (the dated version is archived in `report/archive`).

2026-11-01 Λ-style proton PID cut adopted (B 4.56 → 2.48, limit 9.3 → 8.1× GENIE at $f = 15\%$); the PeLEE/Λ-style crossover moves to $f \approx 21\%$. Muon-track photon veto, loosened cuts and a no-shower second bin measured and not used.